

DK LEARNING

GRADE 8

Python & Data Logic

Level: **Advanced+** • Purpose: Learn before you practise

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In this guide

1. Lists
2. Looping Through Lists
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5. Solving Data Problems
6. Practice questions, answers and glossary



How to use this guide

Read each chapter slowly. Look at the example, then try the “Your turn” task **without copying**. Write your own notes on the lines. When you finish, answer the practice questions and check your work with the answer page. After that, open the Grade 8 activity on the website.

Learning goals

- Store several values in lists
- Write and use functions
- Combine conditions with Boolean logic
- Process data with loops

Study plan

Step	What to do
1	Read the chapter and highlight key words
2	Study the worked example
3	Do the “Your turn” task on paper or on a computer
4	Write a short summary in your own words
5	Answer the practice questions, then check

Tip: When your code gives an error, read the last line of the message first. It usually tells you what is wrong.

1 Lists

A list stores multiple values in an ordered collection. Python lists can contain numbers, strings or other values. Each item has an index that starts at 0.

You can add items with `append()`, remove with `remove()` and find the length with `len()`.

Key points

- Created with []
- Index starts at 0
- `append()` adds an item
- `len()` counts items

Worked example

```
marks = [72, 55, 91, 40, 68]
print(len(marks))      # 5
print(marks[0])        # 72
marks.append(83)
```

Your turn

Create a list of five marks and print the first and last item.

Check yourself

Explain each term in your own words:

List

Index

My notes

2 Looping Through Lists

A for loop can visit every item in a list. This helps you add numbers, find the highest value or count items that meet a rule.

Use a variable such as total to collect results while looping.

Key points

- for item in list
- Use a counter or total
- Built-in: sum(), max(), min()
- Check the loop result

Worked example

```
marks = [72, 55, 91, 40, 68]
total = 0
for m in marks:
    total = total + m
print(total / len(marks))    # average
```

Your turn

Count how many marks in the list are 50 or more.

Check yourself

Explain each term in your own words:

Accumulator

Average

My notes

3 Functions

Functions package reusable instructions. A function can receive inputs (parameters) and return a result. Use `def` to create one.

Functions make programs shorter, tidier and easier to test.

Key points

- `def` creates a function
- Parameters are inputs
- `return` gives back a result
- Call it by its name

Worked example

```
def add(a, b):  
    return a + b  
  
print(add(4, 7))    # 11
```

Your turn

Write a function `square(n)` that returns `n` multiplied by itself.

Check yourself

Explain each term in your own words:

Parameter

Return value

My notes

4 Boolean Logic

Boolean expressions are True or False. Operators such as and, or and not help combine conditions. and needs both to be True, or needs at least one, and not reverses the value.

Truth tables list all possible results and help you check logic.

Key points

- and = both True
- or = at least one True
- not = opposite
- Truth tables show every case

Worked example

```
age = 15
has_id = True
if age >= 13 and has_id:
    print("Allowed")
```

Your turn

Complete a truth table for A and B using True/False.

Check yourself

Explain each term in your own words:

Boolean

Truth table

My notes

5 Solving Data Problems

To solve a problem: understand it, plan steps, write code, test and improve. Use sample data to test normal values and unusual values.

Break big problems into smaller functions and test each one.

Key points

- Understand, plan, code, test
- Test with different data
- Break problems into parts
- Comment your code

Worked example

```
def grade(mark):  
    if mark >= 75: return "A"  
    elif mark >= 50: return "S"  
    return "F"  
print(grade(82))
```

Your turn

Write a function that takes a list of marks and returns the highest mark.

Check yourself

Explain each term in your own words:

Test case

Comment

My notes



Practice Questions

Part A - Choose the best answer

1. What is the index of the first list item?

- A) 0
- B) 1
- C) -1

2. Which keyword creates a function?

- A) def
- B) func
- C) make

3. not True is...

- A) False
- B) True
- C) 0

4. Which adds an item to a list?

- A) append()
- B) add_all()
- C) plus

5. len([4,5,6]) is...

- A) 3
- B) 2
- C) 6

6. $x > 3$ and $x < 8$ is True when x is...

- A) 5
- B) 9
- C) 2

Part B - Short answers

1. What does return do?

2. Why use functions?

3. What is list[2]?

4. What does or mean?



Answer Page

Part A

1-A 2-A 3-A 4-A 5-A 6-A

Part B (sample answers)

1. Gives a result back from a function.
2. To reuse code and keep programs tidy.
3. The third item in the list.
4. True if at least one condition is True.

Mini glossary

List	An ordered collection of values
Index	The position number of an item
Accumulator	A variable that collects a running total
Average	Total divided by the number of items
Parameter	A value a function receives
Return value	The result a function gives back
Boolean	A value that is True or False
Truth table	A table of all input and output combinations
Test case	Sample input used to check code
Comment	A note in code ignored by Python



Activity Preparation

Complete data-logic questions and write a Python function or loop solution. Then open the Grade 8 activity.

Think before you start

- Explain one important idea from this grade in your own words.
- Give one real-life example where this topic is useful.
- What mistake could a beginner make, and how can it be prevented?
- Write the steps you will follow before starting the Grade 8 activity.

Ready checklist

- | | |
|---|--|
| ☐ I read every chapter | ☐ I tried the “Your turn” tasks |
| ☐ I understand the key terms | ☐ I checked my practice answers |
| ☐ I wrote my own notes | ☐ I am ready for the activity |

⚡ **START GRADE 8 ACTIVITY**

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